SOLVANE DYNAMICS INC.

Solvane FleetOS — Product Specification, v3.1

This document describes a fictional company, fictional employees, and fictional financial data. It was prepared for a retrieval-augmented generation (RAG) exercise.

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Contents

Section numbers are stable anchors used for retrieval and citation.

1. Product overview

Solvane FleetOS is Solvane Dynamics' fleet-orchestration and analytics platform for autonomous mobile robots (AMRs) deployed in third-party logistics warehouses. FleetOS coordinates robot task assignment, charging schedules, and traffic management, and surfaces throughput analytics to warehouse operations managers. FleetOS reached general availability in Q2 of the prior fiscal year and now runs in 34 warehouse sites across 11 customers as of the end of Q3 of the current fiscal year.

2. Target customers and market

FleetOS is designed for third-party logistics (3PL) operators and mid-size e-commerce fulfillment centers running 15–200 AMR units per site. Solvane does not manufacture robots exclusively for its own platform; FleetOS supports mixed fleets from multiple hardware vendors, provided the vendor implements the Solvane Fleet Interface (SFI) protocol.

3. Competitive landscape

FleetOS's primary competitors are vertically-integrated robotics vendors (which lock customers into single-vendor hardware) and legacy warehouse execution systems that were not designed for real-time multi-robot traffic coordination. Solvane's differentiation is vendor-neutral orchestration: a customer can mix AMR hardware from different manufacturers on the same floor under one FleetOS instance.

4. Core features

Dynamic Task Assignment: assigns pick, restock, and transport tasks to the nearest available robot using a real-time cost model.

Traffic & Congestion Management: prevents robot gridlock in narrow aisles using a reservation-based routing protocol.

Charge Scheduling: predicts fleet charging needs and schedules charging during low-task-volume windows to maximize uptime.

Throughput Analytics Dashboard: tracks picks-per-hour, robot utilization, and idle-time root causes by zone.

Exception Alerts: notifies floor supervisors of stalled robots, blocked charging docks, or tasks exceeding an expected time threshold.

Multi-Vendor Fleet Support: coordinates robots from different hardware vendors under a single orchestration layer via the SFI protocol.

5. Integrations

FleetOS integrates with warehouse management systems (WMS) including Manhattan Active and Blue Yonder WMS via a published REST API, and ingests task queues over MQTT for lower-latency deployments. A public analytics API allows customers to pull throughput data into their own BI tools.

6. Technical architecture

FleetOS runs as a set of microservices deployed on AWS (primary region us-west-2, disaster-recovery region us-east-1). The task-assignment and traffic-coordination services are written in Go for low-latency real-time decisions; the analytics and dashboard services are Python (FastAPI) with a React front end. Robot telemetry is ingested via MQTT into a time-series store (TimescaleDB); operational and customer configuration data is stored in PostgreSQL.

7. Security and compliance

FleetOS completed a SOC 2 Type II audit in the prior fiscal year with two minor findings, both remediated within 30 days. Customer facility layout and telemetry data are encrypted at rest (AES-256) and in transit (TLS 1.2+). Data retention for robot telemetry defaults to 18 months unless a customer requests a shorter window.

8. Service level agreements

Tier	Orchestration uptime SLA	Support response
Standard	99.5%	Next business day
Priority	99.9%	4 business hours
Enterprise	99.95%	1 hour, 24/7

9. Pricing tiers

Tier	Monthly price	Robot limit	Support
Standard	$2,200	Up to 25 robots	Business hours email
Priority	$5,800	Up to 100 robots	Email + chat, 4h SLA
Enterprise	Custom	Unlimited	Dedicated CSM, 1h SLA

10. Release history

Q2 (prior FY): General availability launch with Dynamic Task Assignment and Traffic Management. **Q4 (prior FY):** Added Charge Scheduling. **Q2 (current FY):** Added Multi-Vendor Fleet Support via the SFI protocol. **Q3 (current FY):** Added Exception Alerts and expanded WMS integration to Blue Yonder.

11. Q4 roadmap (planned, not committed)

The product team has proposed, but not finalized, a Q4 roadmap that includes predictive maintenance alerts (flagging robots likely to fail before a breakdown occurs) and a customer-facing SLA compliance report. Formal Q4 revenue projections have not been published as of this document's release and will follow the Q4 planning offsite.

12. Known limitations

FleetOS does not currently support outdoor or cross-dock yard robots, only indoor warehouse-floor AMRs. Traffic coordination accuracy degrades in facilities with more than 40% of aisles narrower than 1.2 meters; Solvane recommends a site survey before deployment in older, narrow-aisle warehouses.

13. Glossary

AMR: Autonomous Mobile Robot. **SFI**: Solvane Fleet Interface, the vendor-neutral protocol robots implement to be orchestrated by FleetOS. **WMS**: Warehouse Management System, the customer's existing inventory/order system that FleetOS integrates with rather than replaces.